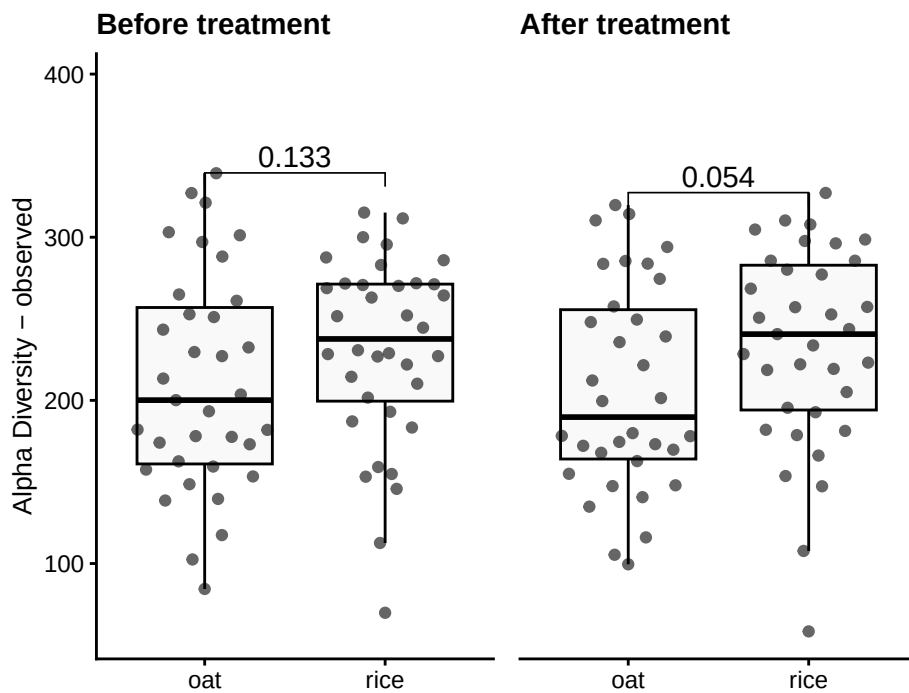


Alpha Diversity

Muluh

1 Group-wise comparisons

- **Index:** observed
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



1.1 Table of the top findings

1.1.1 Across-group

1.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	210.9	231.3	0.13	0.133

Before treatment, the rice group had a slightly higher observed alpha diversity (231.30) compared to the oat group (210.90). The fold change (log2FC) of 0.13 suggests a small increase in richness for the rice group. However, the corrected p-value ($p_{adj} = 0.133$) indicates that this difference is not statistically significant, meaning it could be due to random variation rather than a true effect of diet.

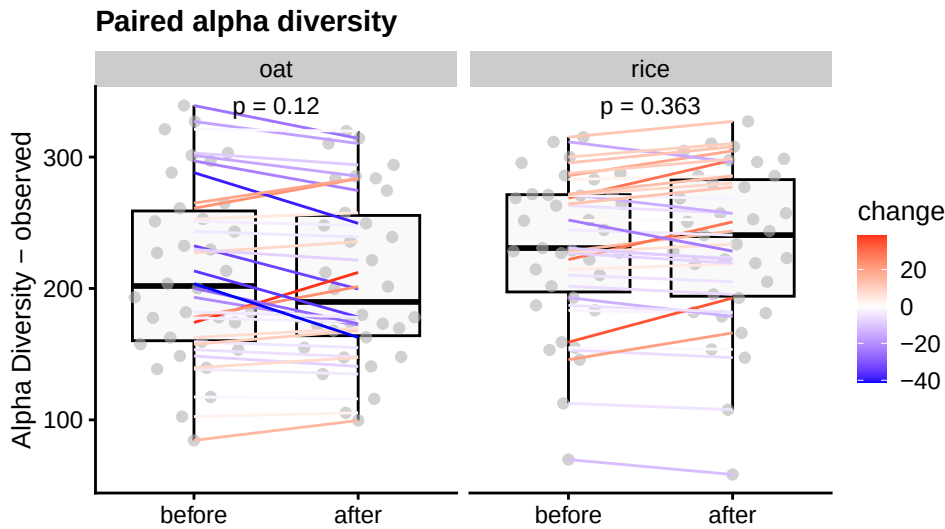
1.1.1.2 After treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	206.9	233	0.17	0.054

After treatment, the rice group (233.00) continued to show a higher observed alpha diversity compared to the oat group (206.90), with a fold change (log2FC) of 0.17. The adjusted p-value (p.adj = 0.0540) is approaching statistical significance, suggesting a potential effect of diet on diversity, though it remains marginal.



1.1.2 Within-group

1.1.2.1 Paired treatment

- Multiple testing correction: **paired -none**
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
oat	210.9	206.9	0.04	0.120
rice	231.3	233.0	0.13	0.363

Oat group: Observed alpha diversity slightly decreased from 210.90 to 206.90 (log2FC = 0.04, p = 0.120), indicating a small but non-significant reduction over time.

Rice group: Diversity increased modestly from 231.30 to 233.00 (log2FC = 0.13, p = 0.363), but this change is also not statistically significant.

Overall, there were no significant within-group changes in observed alpha diversity over time, suggesting that dietary intervention had a minimal effect on diversity shifts within each group.

1.2 LMM

- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	151	67, 235	<0.001	144	59, 229	0.001
meal_group			0.4			0.3
1	—	—		—	—	
2	32	-11, 75	0.14	28	-16, 72	0.2
3	28	-15, 70	0.2	30	-14, 73	0.2
4	10	-32, 53	0.6	-3.8	-47, 39	0.9
age	1.0	-0.44, 2.5	0.2	1.2	-0.25, 2.7	0.10

¹CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI ¹	p-value
(Intercept)	147	84, 210	<0.001
meal_group			0.6
1	—	—	
2	28	-16, 71	0.2
3	28	-15, 71	0.2
4	2.2	-27, 32	0.9
age	1.2	0.13, 2.2	0.026

¹CI = Confidence Interval

- Formula used: `~ diet * timepoint + (1 | id)`

Table 4: Table of interaction effects

Term	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	7.672	0.084	88.336	90.944	0.000
dietrice	0.115	0.118	88.333	0.973	0.333
time	-0.027	0.020	67.103	-1.365	0.177
dietrice:time	0.036	0.028	67.102	1.291	0.201

Diet effect (rice vs. oat): No significant difference between rice and oat groups (**Estimate** = 0.12, **p** = 0.333), suggesting baseline diversity was similar.

Time effect: A slight decrease in diversity over time (**Estimate** = -0.03, **p** = 0.177), but this was not statistically significant.

Diet × Time interaction: The interaction term (**Estimate** = 0.04, **p** = 0.201) suggests that the change in diversity over time did not significantly differ between the oat and rice groups.

2 Alternate analyses

- **Beta (Estimate):** Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI:** Range where the true beta is likely to fall, reflecting precision.
- **p-Value:** Tests if beta differs from zero; $p < 0.05$ suggests significance.

3 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(16): sample id ... group observed_log2
reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```